

5-Mark Questions & Answers — Data Structures Module 6: Sorting & Searching Techniques

Q1. Explain Bubble Sort, Insertion Sort, and Selection Sort with step-by-step trace and C programs.

Detailed algorithms, code implementations, and complexity analysis ($O(N^2)$ average, $O(1)$ space).

Q2. Explain Merge Sort and Quick Sort algorithms with Divide-and-Conquer recurrence relations.

- **Merge Sort:** Divide in half, recursively sort, merge halves. $T(N) = 2T(N/2) + O(N)$ implies $O(N \log N)$ in all cases.
 - **Quick Sort:** Partition around pivot. $O(N \log N)$ average, $O(N^2)$ worst case. In-place sorting.
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Q3. Explain Collision Resolution Techniques in Hashing with numerical examples.

Detailed walkthrough of Linear Probing, Quadratic Probing, Double Hashing, and Separate Chaining.
